

PA-VM

DASHBOARD

ACC

MONITOR

POLICIES

OBJECTS

NETWORK

DEVICE

Security

NAT

QoS

Policy Based Forwarding

Decryption

Tunnel Inspection

Application Override

Authentication

DoS Protection

SD-WAN

NAME

TAGS

TYPE

ZONE

ADDRESS

USER

DEVICE

1	intrazone-default	none	intrazone	any	any	any	any
2	interzone-default	none	interzone	any	any	any	any

Policy Optimizer

New App Viewer

Rules Without App Controls

Unused Apps

Rule Usage

Unused in 30 days

Unused in 90 days

Unused

Object : Addresses

Add

Delete

Clone

Override

Revert

Enable

Disable

Move

PDF/CSV

Highlight Unused

	NAME	TAGS	TYPE	Source				Destination			APPLICATION	SERVICE
				ZONE	ADDRESS	USER	DEVICE	ZONE	ADDRESS	DEVICE		
1	intrazone-default	none	intrazone	any	any	any	any	(intrazone)	any	any	any	any
2	interzone-default	none	interzone	any	any	any	any	any	any	any	any	any

Security Policy Rule

General

Source

Destination

Application

Service/URL Category

Actions

Name

BlockingApplication

Rule Type

universal (default)

Description

Tags

Group Rules By Tag

None

Audit Comment

[Audit Comment Archive](#)

OK

Cancel

Then go to Source tab → select “any”:

Security Policy Rule

General | **Source** | Destination | Application | Service/URL Category | Actions

☒ Any	☒ Any	any	any
☐ SOURCE ZONE ^	☐ SOURCE ADDRESS ^	☐ SOURCE USER ^	☐ SOURCE DEVICE ^
+ Add - Delete	+ Add - Delete	+ Add - Delete	+ Add - Delete

☐ Negate

OK Cancel

Then to ‘destination’

Security Policy Rule

General | Source | **Destination** | Application | Service/URL Category | Actions

any	☒ Any	any
☐ DESTINATION ZONE ^	☐ DESTINATION ADDRESS ^	☐ DESTINATION DEVICE ^
+ Add - Delete	+ Add - Delete	+ Add - Delete

☐ Negate

OK Cancel

Now go to Application → Add → add required application like Facebook, YouTube and so on.

Security Policy Rule

General | Source | Destination | **Application** | Service/URL Category | Actions

☐ Any	7 items → ×
☐ APPLICATIONS ^	DEPENDS ON ^
☐ facebook ☒ youtube	☐ google-base ☐ mqtt-base ☐ rtcp ☐ rtp-base ☐ ssl ☐ stun ☐ web-browsing
+ Add - Delete	Add To Current Rule Add To Existing Rule

OK Cancel

Now go to Service/URL Category tab:

The screenshot shows the 'Security Policy Rule' configuration window with the 'Service/URL Category' tab selected. The 'application-default' dropdown is highlighted. The 'SERVICE' and 'URL CATEGORY' sections are empty, each with an 'Add' button and a 'Delete' button. The 'Any' checkbox is checked under the 'URL CATEGORY' section. The 'OK' and 'Cancel' buttons are at the bottom right.

Then go to Actions tab:

The screenshot shows the 'Security Policy Rule' configuration window with the 'Actions' tab selected. The 'Action Setting' section has 'Deny' selected. The 'Log Setting' section has 'Log at Session Start' and 'Log at Session End' checked. The 'Profile Setting' section has 'None' selected. The 'Other Settings' section has 'Schedule' and 'QoS Marking' set to 'None'. The 'OK' and 'Cancel' buttons are at the bottom right.

The screenshot shows the PA-VM dashboard with the 'Security' tab selected. The table displays the configuration for the 'BlockingApplication' rule.

NAME	TAGS	TYPE	ZONE	ADDRESS	USER	DEVICE	ZONE	ADDRESS	DEVICE	APPLICATION	SERVICE	ACTION	PROFILE	OPTIONS
BlockingApplication	none	universal	any	any	any	any	any	any	any	facebook	application...	Deny	none	
Intrazone-default	none	intrazone	any	any	any	any	(intrazone)	any	any	any	any	Allow	none	none
Interzone-default	none	interzone	any	any	any	any	any	any	any	any	any	Deny	none	none

Then commit the configuration.

The screenshot shows the 'Commit Status' dialog box. The 'Operation' is 'Commit', the 'Status' is 'Active', and the 'Result' is 'Pending'. The progress bar is at 0%. The 'Commit' button is highlighted.

The screenshot shows the 'Commit Status' dialog box after the configuration has been committed successfully. The 'Operation' is 'Commit', the 'Status' is 'Completed', and the 'Result' is 'Successful'. The details show 'Configuration committed successfully'. The 'Commit' button is highlighted.

To validate the configuration → go to client VM and access sites like Facebook & YouTube.

Once you have procured license, then only App Id practical can be done.

URL Filtering advanced

URL Filtering in a Palo Alto Networks firewall is a **Content-ID security control** that allows administrators to **classify, monitor, allow, block, alert, or override web traffic** based on the **category of the requested URL**, rather than relying solely on IP addresses or ports.

Palo Alto uses **PAN-DB (URL Filtering Database)** or **BrightCloud** (legacy/optional) to dynamically categorize URLs in real time.

Key Characteristics

- Works at **Layer 7 (Application Layer)**
- Enforced **within Security Policies**
- Uses **App-ID + Content-ID**
- Can control:
 - Malicious URLs
 - Productivity-impacting sites
 - Legal/compliance-restricted categories
- Supports **User-ID based filtering**

URL Filtering Categories (Examples)

Common PAN-DB categories include:

- malware
- phishing
- command-and-control
- adult
- social-networking
- streaming-media
- online-storage-and-backup
- hacking
- gambling
- unknown

Prerequisites Before Configuration

Ensure the following are in place:

1. **Valid URL Filtering License**
 - Device → Licenses → URL Filtering
2. **Security Policy allowing web traffic**
3. **DNS resolution working**
4. (Optional but recommended) **SSL Decryption enabled**
5. User-ID configured if user-based control is required

Step-by-Step: Configure URL Filtering in Palo Alto Firewall

Step 1: Create a URL Filtering Profile

1. Go to
Objects → Security Profiles → URL Filtering
2. Click **Add**
3. Name the profile
Example: URLF-Internet-Policy

Step 2: Configure Category Actions

Category	Recommended Action
malware	block
phishing	block
command-and-control	block
adult	block
hacking	block
gambling	block
social-networking	alert / allow
streaming-media	alert / block
unknown	block (strict environments)
low-risk	allow

Step 3: Configure Advanced Settings (Recommended)

Under the **Settings** tab:

- Enable **Safe Search Enforcement**
- Enable **Log Container Page Only**
- Enable **User Credential Detection**
- Enable **Block Credential Phishing**

Step 4: Attach URL Filtering Profile to a Security Policy

1. Navigate to
Policies → Security
2. Edit the **Internet Access** policy rule
3. Go to the **Actions** tab
4. Under **Profile Setting**:
 - Select **Profiles**
 - Set **URL Filtering** = URLF-Internet-Policy
5. Ensure:
 - Action = **Allow**
 - Application = web-browsing, ssl

URL Filtering does **not work on deny rules**.

Step 5: Commit the Configuration

- Click **Commit**
- Verify commit success

Step 6. Step-by-Step Validation

Test from Client Machine

- Try accessing:
 - <http://www.facebook.com>
 - <http://testphishing.com>
 - <http://exampleadultsite.com>

Check Logs

1. Go to
Monitor → URL Filtering
2. Verify:
 - URL
 - Category
 - Action (block/allow/alert)
 - User (if User-ID enabled)